How Did Revolutionary Cadre Deployment Shape Regional Economic Inequality in Mid-20th Century China?

Word count: 3,329

Introduction

In the past decades, institution has triggered rich discussion as the main economic framework in investigating economic disparity among countries, however, less literatures investigates the domestic regional economic disparity in China. While China's economic development has been considered "miracle" for its non-democratic political institution, main studies attributes them to differences in endowments, such as physical capital (Xu Xianxiang & Shu Yuan, 2004), human capital (Cai Fang & Du Yang, 2000), geographic location (Bao et al., 2002), and infrastructure (Démurger et al., 2002); or emphasizes the role of external interventions, such as central government's preferential policies (Fleisher & Chen, 1997), FDI (Lee, 1994), and development strategies (Lin Yifu & Liu Peilin, 2003). However, these factors are often the result of policies rather than their cause, and cannot fully explain internal regional differences, nor can they account for the changing trends in regional disparities.

Recent scholarship has turned attention to the role of de facto political power—the actual ability of local governments to influence resource allocation and policy outcomes, beyond what is formally authorized. Works by Huang (1996), Landry (2008), and Sun et al. (2022) emphasize the important role of de facto political power of local government in economic policies and resource allocation. The historical revolutionary war has affected the structure of local cadre deployment in the 1950s-1960s in China, and has resulted in "China federalism" that influence the de facto political power of local cadre (Montinola, Qian and Weingast, 1995). This contributes to explain the disparity in regional economic development in China. This paper investigates whether the composition of local cadres in the provincial government—many of whom were embedded in their native regions compared to whom from the central government—has lasting effects on regional economic performance.

This study contributes to the literature by examine the relationship of institutional historical heritage to cadre localization and de facto political power, hence to explain China's regional economic disparity. Drawing from frameworks proposed by Acemoglu and Robinson, as well as the Chinese literature on cadre management and state decentralization, this paper argues cadre structure as a durable form of informal authority. The theoretical expectation is that higher levels of cadre localization reduce administrative neutrality, weaken vertical coordination, and diminish developmental state capacity, and ultimately harming local economic outcomes.

To examine this hypothesis, this paper uses original data on cadre origin ratios from the 1950s

and matches them with economic indicators from 1964 provincial statistical records. Using a stepwise OLS regression framework, it finds that provinces with higher proportions of locally born cadres had lower log GDP per capita.

The remainder of the paper is structured as follows: Section 2 reviews the literature on institutional persistence, de facto power, and regional inequality; Section 3 introduces the data and methodology; Section 4 presents regression results and interpretation; Section 5 discusses robustness checks and theoretical implications; and Section 6 concludes the findings.

Literature review

After 1949, China's revolutionary war and early state-building left a clear institutional legacy. This variation in cadre composition was not random; it closely reflected the distribution of _de facto_ political power during the war period. Historically, the proportion of locally born cadres in the provincial leadership was closely affected by the lasting effects of the Civil War. In areas where revolutionary bases were established during the War, the share of local cadres was usually high. As shown in Figure 1, provinces such as Hebei, Shandong, those of which had revolutionary bases, had local cadre shares above 80%. In contrast, places that were under the Chinese Nationalist Party's control or more isolated, such as coastal regions, inland provinces, and areas like Yunnan and Guizhou , had zero local cadres in their first committees.

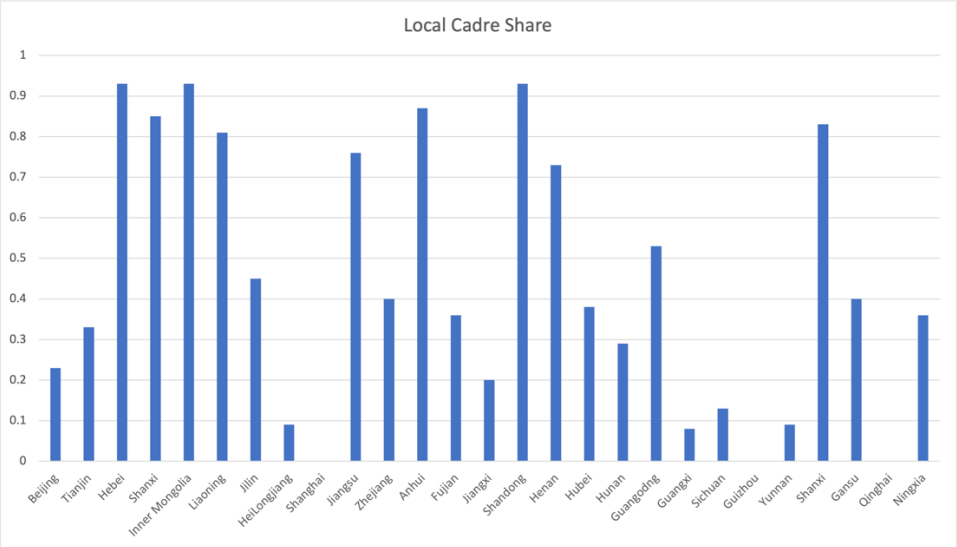


Figure 1 The Proportion of Local Cadre in 1st Provincial Government 1956
Data Source:

Importantly, these initial power structures did not disappear with institutional changes, but they tended to reproduce themselves over time. In areas with revolutionary bases, local cadres had gone through wartime and built strong organizational and military capacity, as well as gained trust from the central government. During the Civil War, local armed forces could defense themselves, and locals mostly managed the new governments. In contrast, areas without such a

history suffered from war by external armed forces, and their governments were taken over by cadres from outside. The structure of the first provincial committee that was formed through political bargaining reflected the distribution of grassroots political power. A higher share of local cadres meant that local grassroots forces held more de facto political power.

More importantly, this initial power structure tended to persist and strengthen itself over time. In China's political system, personal networks play a key role in promotion. When local cadres dominate, they often promote those with similar backgrounds or prior work connections. These ties can be extended into family or social relations, such as kinship, marriage, hometown, or school networks. These informal networks then expand in both depth and scope and can even be passed down across generations. This self-reinforcing mechanism makes the original structure of de facto power more stable, although increasingly hidden and complex.

Empirical work by Landry (2008) shows how the CCP institutionalized cadre localization through a "one-level-down" promotion system, which allowed local Party committees to appoint officials within their own jurisdictions. This reinforced the dominance of local networks and made vertical supervision from the center more difficult. Although China is officially a unitary state, such arrangements produced a system of "decentralized authoritarianism," where local officials exercised considerable de facto power. Later studies confirm the persistence of this structure. Landry and Liu (2011) find that even in the reform era, provincial leaders still showed strong home-origin preferences in cadre selection. Edin (2003) similarly finds that local cadre management remained embedded in personal ties and faced ongoing principal–agent problems. These findings align with the idea that revolutionary-era political arrangements, created long-lasting informal institutions that continued to shape governance styles and regional policy outcomes. Together, this literature suggests that cadre localization is both a product of wartime politics and a key mechanism through which de facto political power endures.

Beyond political structure, cadre localization also has important implications for regional economic development. Important consequences for regional development. Mahoney (2000) argues that institutional "lock-in" happens when initial arrangements create self-reinforcing paths over time. In China, cadre localization may lead to governance practices that favor administrative stability and political loyalty, rather than economic experimentation or coordination across provinces. Huang (1996) notes that during the reform era, subnational governments gained increasing control over economic affairs, which made the background and incentives of local officials more important. When resource allocation and policy implementation depend more on informal local networks than on central plans, the composition of local elites becomes a key factor. Recent empirical studies support this view. Sun et al. (2022) show that de facto political power is a stronger predictor of resource outcomes than de jure rules in transitional China. Ang (2016) finds that clientelist networks, not formal institutions, often determine who gets access to development resources. Similarly, Su and Heurlin (2019) argue that in authoritarian systems, local responsiveness comes more from informal flexibility than from merit-based rules.

Despite these theoretical and empirical insights, one question remains unclear: How has

revolutionary-era cadre localization shaped long-term regional economic outcomes? Scholars like Montinola, Qian, and Weingast (1995) describe "Chinese federalism" as a system where provinces have policy autonomy under a unified political structure. Others, such as Xu (2011), emphasize how regional competition drives growth. However, few works empirically connect early cadre deployment patterns to differences in regional development in later decades. Likewise, studies of state-building (Slater 2010; Sinha 2007) highlight how early institutional choices shape later outcomes, but they often ignore how the composition of local governing elites influences these paths in authoritarian settings like China.

This paper aims to fill that gap by testing whether the share of locally born cadres in provincial governments, as a proxy for de facto power, has a long-term effect on economic performance. The paper argues that cadre localization weakens vertical coordination, reduces the incentives for growth-focused governance, and contributes to regional inequality. By linking the cadre structure of the 1950s to GDP outcomes in the 1960s, the study provides historical evidence for how revolutionary institutions continue to shape China's political economy today.

Data and Methodology

This paper tests whether the structure of political authority (localization of provincial leadership) established in the early years of China, has a lasting impact on regional economic outcomes. The theoretical expectation is that a higher proportion of locally born cadres would weaken vertical coordination, reduce administrative efficiency, and resource allocation, and ultimately harm economic performance. To empirically test this, the study constructs a cross-sectional regression framework using mid-20th century provincial-level data in China.

Key Variables

The main independent variable is cadre ratio, defined as the proportion of provincial Party Standing Committee members in 1956 who were born or were raised in the same province. This measure captures localized de facto political power—an institutional legacy from the revolutionary period.

The dependent variable is log GDP per capita in 1964, constructed from provincial-level industrial and agricultural output data combined with population statistics from the 1964 China Statistical Yearbook. This choice of year allows a temporal gap from cadre assignment, helping reduce reverse causality while still reflecting early state-building outcomes.

To isolate the effect of cadre localization, the following control variables are introduced:

- Industry share: The proportion of industrial output relative to total industrial and agricultural output, capturing the degree of structural transformation.
- Industry-agriculture ratio: A continuous measure of modernization, reflecting the balance between industrial and agricultural output.
- Urban and rural employment shares: Capturing labor structure and local economic

capacity.

- Population total: Used in robustness checks to control for scale effects.

These controls help account for alternative explanations of regional development, particularly structural economic differences.

Model Specification and Strategy

The empirical strategy uses ordinary least squares (OLS) regression. The core equation is:

$$\log(GDP_i) = \beta_0 + \beta_1 CadreRatio + \beta_2 ControlVariables + \epsilon_i$$

Seven regression models (M1–M7) are constructed following a stepwise inclusion strategy, designed to test both robustness and mechanisms. This approach is chosen to isolate the direct association between cadre localization and economic performance, as well as test whether this association is mediated or moderated by structural controls.

- Model 1-2: examine the correlation between cadre ratio and GDP per capita.
- Model 3-4: add employment and industrial structure step by step.
- Model 5-6: transform the dependent variable to logarithmic form and adjust specifications for better fit and robustness.

(See Appendix)

Based on prior literature, the paper expects a negative and significant coefficient on cadre ratio. Provinces with a higher proportion of local cadres—especially those with revolutionary base areas—were more likely to develop personalized governance networks that reduced reform flexibility and hindered coordinated economic initiatives. A positive coefficient on industry share is also expected, as industrialization has been widely recognized as a driver of economic performance in 20th-century China.

Regression Analysis and Interpretation

The empirical analysis supports the hypothesis that a higher proportion of locally born cadres is associated with lower levels of provincial economic development. The main regression (Figure 2) reveals a statistically significant and negative relationship between cadre ratio and log GDP per capita, even after controlling for industrial structure. The coefficient of -0.351 ($p = 0.045$) implies that a one-unit increase in cadre localization is associated with a 35% decrease in GDP per capita, holding industry share constant. This result is visually supported by Figure 3, which shows a downward trend in predicted GDP as cadre ratio increases.

VARIABLES	(1) M1 Baseline Model	(2) M2 Control for Labor Market	(3) M3 Control for Industrial Share	(4) M4 Log GDP per capita
Cadre ratio	-151.921	-347.315**	-148.114**	-0.351**
	(100.847)	(153.629)	(54.052)	(0.166)
Urban employ		1.228*		
		(0.620)		
Rural employ		0.093		
		(0.115)		
Industry share			969.761***	2.898***
			(122.149)	(0.375)
Constant	301.480***	145.909	-38.870	4.445***
	(55.107)	(109.134)	(52.059)	(0.160)
Observations	27	14	27	27
R-squared	0.083	0.503	0.747	0.729

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Note: All regressions are estimated using Ordinary Least Squares (OLS). Robust standard errors are reported in parentheses.
Model (1) is the baseline regression of GDP per capita on Cadre ratio.
Model (2) adds employment structure controls: Urban employ and Rural employ.
Model (3) adds Industry share to the baseline.
Model (4) is the main specification with log-transformed GDP per capita and Industry share.
Model (5) is the log-linear baseline without additional controls.
Model (6) reintroduces employment controls under the log specification.

Figure 2 (continued)

VARIABLES	(5) M5 Logged Baseline	(6) M6 Logged M2
Cadre ratio	-0.362	-1.000*
	(0.304)	(0.450)
Urban employ		0.005**
		(0.002)
Rural employ		-0.000
		(0.000)
Industry share		
Constant	5.463***	5.068***
	(0.166)	(0.319)
Observations	27	14
R-squared	0.054	0.478

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

This finding supports the main argument that, political control often lies not in formal institutions but in embedded networks (Acemoglu & Robinson, 2008; Landry, 2008). In provinces where local cadres dominated the initial political structure (has revolutionary base areas), the local government were able to build informal systems of governance that resisted central oversight and slowed institutional modernization.

In practical terms, these localized networks limited vertical accountability. Local cadres were often evaluated by their immediate superiors—who were also locals—rather than by external actors. As Edin (2003) and Landry (2008) show, this led to widespread principal-agent problems: underperformance hardly led to dismissal, and promotions often favored loyalty over capacity. For example, in post-revolutionary Shanxi and Hebei, provincial leadership remained internally homogenous throughout the 1950s and 60s, resisting the introduction of external reformers.

Additionally, Model 7 that includes both cadre ratio and industry share, explains over 72% of the variance in log GDP per capita ($R^2 = 0.7287$), confirming that both political and economic structures mattered. Figure 4 further confirms the robustness of the cadre ratio effect across model specifications, even as control variables vary. This reinforces the view that cadre composition can largely affect and involve in policy decision making process and resource allocation.

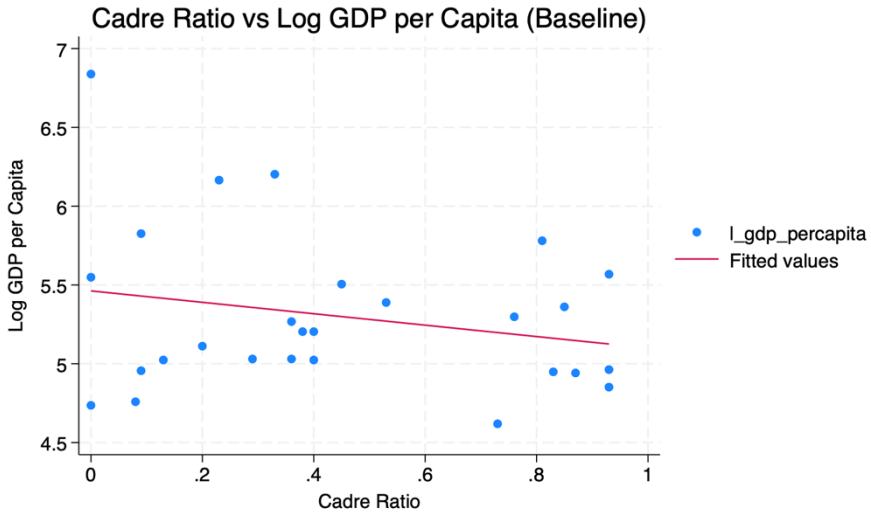


Figure 3 Model 6 Best Fit line

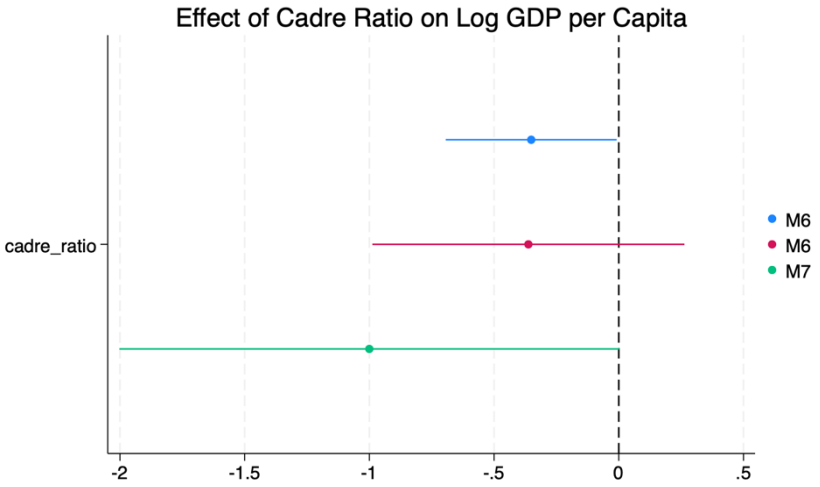


Figure 4 Coefficient Estimates of Cadre Ratio Across Models M1–M6

To further understand this mechanism, it is useful to compare earlier models. Model 1 (Figure 2) includes only cadre ratio and yields an insignificant coefficient. As additional controls are introduced (Model 5 and Model 6), the negative effect becomes statistically significant. This stepwise inclusion helps isolate the independent effect of cadre ratio; Secondly, it tests whether this effect is mediated through economic structure. The results suggest that cadre ratio has an effect independent of industrialization.

This highlights an important mechanism. Informal cadre networks may have locked local governments into rigid administrative styles. Mahoney (2000) describes how early institutional choices can generate self-reinforcing "lock-in" effects. In this case, high levels of cadre localization constrained reform flexibility. For instance, provinces like Guizhou or Yunnan, which had no revolutionary base and were governed primarily by centrally dispatched cadres, were more likely to accept national-level planning initiatives and large-scale industrial cooperation. In contrast, provinces with deeply localized cadres often delayed these efforts in favor of protecting internal networks and resources.

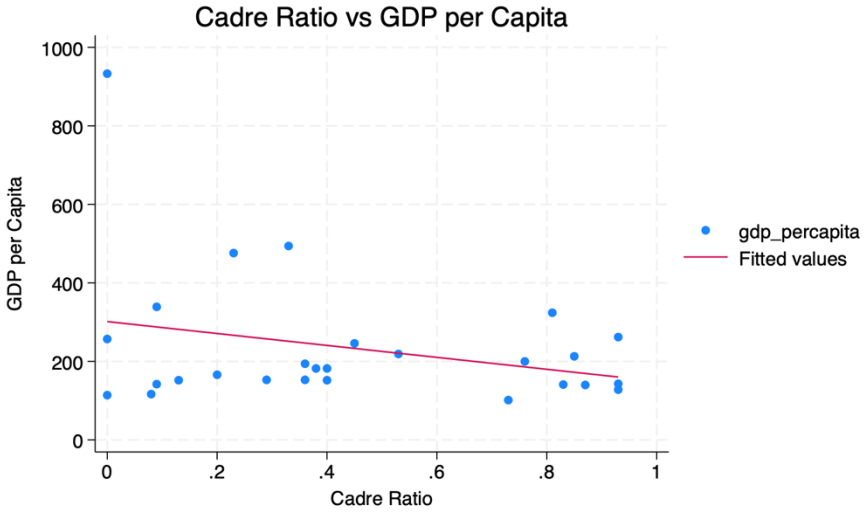


Figure 5 Scatterplot with Fitted Line (M1 Baseline Model)

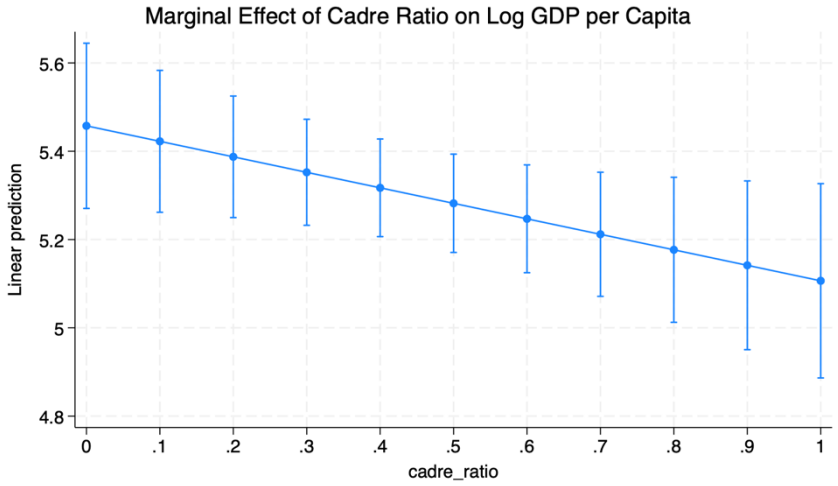


Figure 6 Marginal Effect of M5

Figure 5 illustrates the basic negative correlation between cadre ratio and log GDP per capita in Model 1. While not statistically significant, the scatterplot intuitively supports for the hypothesis. As controls are introduced, Figure 6 shows the marginal predicted GDP value decreasing consistently with cadre ratio, even when other factors are held constant.

Importantly, the control variables behave as expected and support the robustness of the findings.

Industry share is consistently and significantly associated with higher GDP levels ($\beta = 2.90$, $p < 0.001$), confirming the role of structural transformation in economic development. This is consistent with conventional developmental models that stress the importance of industrialization. However, the persistence of cadre ratio's effect even after including industry share highlights that economic structure alone cannot explain interprovincial disparities.

Furthermore, employment structure, introduced in Model 5, provides some predictive power but does not eliminate the significance of cadre ratio. This suggests that while labor market conditions contribute to GDP outcomes, they do not fully mediate the effect of political embeddedness. For example, provinces with high urban employment rates like Jiangsu performed better economically—but only when political coordination capacity was also present.

In sum, the regression results confirm that revolutionary legacies in cadre deployment have had lasting economic effects. Cadre localization, though perhaps efficient for political consolidation in the early PRC, appears to have constrained provincial governments' capacity to coordinate growth-oriented policies and respond to national priorities. These findings provide quantitative backing to theoretical claims of institutional path dependence and highlight the importance of informal power structures in understanding China's uneven development.

Robustness Check and Discussion

Robustness Check

To assess the stability of the findings, multiple model specifications were tested using different combinations of variables and transformations. Across these models from M1 - M6, the coefficient for cadre ratio remains consistently negative and statistically significant in several cases, especially after controlling for key economic variables. For instance, in Model 5 ($p = 0.050$) and Model 6 ($p = 0.045$), cadre ratio has a significant negative association with log GDP per capita. Figure 6 visualizes these effects across all six models, illustrating the stability of the coefficient direction and significance.

Stepwise control variables further supports this robustness. Beginning with labor market variables (urban employment, rural employment) and subsequently adding industry share, the analysis shows that cadre ratio continues to exert an independent negative effect. This suggests that the observed relationship is not merely a proxy for economic structure, but rather reflects a distinct institutional legacy.

Additionally, temporal separation between the independent and dependent variables strengthens the causal interpretation. The use of cadre composition data from 1956 to explain GDP per capita in 1964 ensures that the direction of influence is logically consistent, minimizing risks of reverse causality. This lag structure is crucial for testing institutional persistence and reflects the theoretical expectation that early political structures shape long-

run development trajectories.

Discussion

The empirical findings align closely with theories of institutional path dependence. As argued by Mahoney (2000), localized cadre networks can generate self-reinforcing feedback loops. In the Chinese context, localized cadres promoted successors with similar backgrounds, expanded informal networks through kinship and patronage, and embedded political relationships within administrative power hierarchies. Over time, this led to institutional "lock-in," making reform and coordination more inefficient.

Moreover, the findings highlight the challenges of vertical accountability in decentralized authoritarian systems. The lack of cross-provincial cadre mobility limited oversight and accountability from the center. Local governments, dominated by "native" cadres, prioritized administrative stability and local resource control over growth-oriented experimentation or cross-regional collaboration. This reflects the broader problem of principal–agent failure in authoritarian settings, as emphasized by Acemoglu and Robinson (2008).

Real-world examples underscore these dynamics. Provinces such as Shanxi and Hebei—home to early revolutionary bases and high cadre localization—showed slower industrial coordination and limited reform engagement during the early PRC period. In contrast, provinces like Guizhou, where local cadres were less dominant, were more responsive to centrally coordinated economic strategies. These examples suggest that de facto political power shaped not only bureaucratic behavior but also economic outcomes.

Conclusion

This paper examines the long-term economic implications of revolutionary cadre structures in post-1949 China, focusing on the extent of cadre localization in provincial leadership. By linking original archival data on cadre origins with 1964 provincial economic indicators, it finds a robust and negative association between the proportion of locally born cadres and GDP per capita. This relationship holds even after controlling for industrial structure and employment composition, suggesting that localized cadre systems that was affected by historical factors (revolutionary war) can result in weak institution and can exert persistent influence on economic outcomes.

These findings contribute to the literature on institutional persistence and de facto political power. Building on the work of Acemoglu and Robinson (2008), the analysis highlights how informal political structures, once embedded, may constrain vertical accountability and limit the effectiveness of economic governance. In the Chinese context, localized promotion networks reduced incentives for reform and hindered coordination with central priorities, shaping regional disparities that continued into the reform era.

From a policy perspective, the results suggest potential costs of excessive localization in elite recruitment. It risked self-reinforcing networks that prioritized internal stability over adaptive governance. Cadre rotation, cross-provincial appointments, and strengthened evaluation mechanisms may be necessary to offset the long-run effects of such institutional legacies.

Future research could explore whether these patterns persist beyond the 1960s or compare China's experience with other post-revolutionary or authoritarian states. A longitudinal approach incorporating later reform-era data may also improves on studying how evolving institutional constraints interact with political and economic change. By grounding its analysis in both theoretical debates and historical evidence, this study offers a new perspective on how revolutionary origins continue to shape the political economy of China's regions today.

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Appendix

Appendix A

This appendix summarizes the six main regression models used in the empirical analysis to examine the relationship between cadre localization and regional economic performance in mid-20th century China. All models are estimated using Ordinary Least Squares (OLS). The dependent variable is either GDP per capita or its natural logarithm in 1964. The key independent variable is the cadre ratio, defined as the proportion of provincial cadres in 1956 who were locally born.

Model 1: Baseline Model

$$GDP_{per\ Capita} = \beta_0 + \beta_1 CadreRatio + \epsilon$$

To test the bivariate relationship between cadre localization and GDP per capita without any control variables. This model provides a baseline estimate of the direct correlation.

Model 2 : Control for Labor Market Structure

$$GDP_{per\ Capita} = \beta_0 + \beta_1 CadreRatio + \beta_2 UrbanEmployment + \beta_3 RuralEmploy + \epsilon$$

To control for the composition of the labor force. Urban and rural employment shares may reflect modernization or structural changes affecting GDP.

Model 3 (M3): Control for Industrial Structure

$$GDP_{per\ Capita} = \beta_0 + \beta_1 CadreRatio + \beta_4 IndustryShare + \epsilon$$

To isolate the effect of industrialization by including the share of industrial output relative to total industrial and agricultural production.

Model 4 : Log-Transformed GDP + Industry Share

$$Log[GDP_{per\ Capita}] = \beta_0 + \beta_1 CadreRatio + \beta_4 IndustryShare + \epsilon$$

To assess whether the relationship holds under a log-linear transformation, which accounts for skewed GDP distribution and focuses on proportional changes.

Model 5 : Log Baseline Without Controls

$$Log[GDP_{per\ Capita}] = \beta_0 + \beta_1 CadreRatio + \epsilon$$

To replicate Model 1 using log-transformed GDP, helping to assess whether the basic relationship is sensitive to functional form.

Model 6: Full Log Model with Labor Market Controls

$$Log[GDP_{per\ Capita}] = \beta_0 + \beta_1 CadreRatio + \beta_2 UrbanEmploy + \beta_3 RuralEmploy + \epsilon$$

To extend the log-linear analysis by reintroducing labor market controls under the transformed specification, examining robustness and interaction effects.

Appendix B

The Stata commands are as follows:

```
. pwd
/Users/liujiatong/Desktop/ESPS0015 final

. reg gdp_percapita cadre_ratio

      Source |       SS       df       MS      Number of obs   =        27
-------------+------------------------------   F(1, 25)        =      2.27
       Model |  63797.204      1  63797.204   Prob > F        =    0.1445
    Residual |  702800.902     25  28112.0361  R-squared       =    0.0832
-------------+------------------------------   Adj R-squared   =    0.0466
       Total |  766598.106     26  29484.5425  Root MSE        =    167.67

------------------------------------------------------------------------------
gdp_percap~a | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
 cadre_ratio |  -151.9214   100.8473    -1.51   0.144    -359.6203    55.77751
       _cons |      301.48   55.10663     5.47   0.000     187.9858    414.9742
```

```
------------------------------------------------------------------------------

. outreg2 using regression_results.doc, replace ctitle("M1: Baseline") dec(3)
command outreg2 is unrecognized
r(199);

. ssc install outreg2, replace
checking outreg2 consistency and verifying not already installed...
installing into /Users/liujiatong/Library/Application Support/Stata/ado/plus/...
installation complete.

. reg gdp_percapita cadre_ratio

      Source |       SS       df       MS      Number of obs   =        27
-------------+------------------------------   F(1, 25)        =      2.27
       Model |  63797.204      1  63797.204   Prob > F        =    0.1445
    Residual |  702800.902     25  28112.0361   R-squared       =    0.0832
-------------+------------------------------   Adj R-squared   =    0.0466
       Total |  766598.106     26  29484.5425   Root MSE        =    167.67

------------------------------------------------------------------------------
gdp_percap~a | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
 cadre_ratio |  -151.9214   100.8473    -1.51   0.144    -359.6203    55.77751
       _cons |      301.48   55.10663     5.47   0.000     187.9858    414.9742
------------------------------------------------------------------------------

. outreg2 using regression_results.doc, replace ctitle("M1: Baseline") dec(3)
/Users/liujiatong/Library/Application Support/Stata/ado/plus/o/outreg2.ado
regression_results.doc
dir : seeout

. reg gdp_percapita cadre_ratio urban_employ rural_employ

      Source |       SS       df       MS      Number of obs   =        14
-------------+------------------------------   F(3, 10)        =      3.37
       Model |  316602.985     3  105534.328   Prob > F        =    0.0627
    Residual |  312797.508     10  31279.7508   R-squared       =    0.5030
-------------+------------------------------   Adj R-squared   =    0.3539
       Total |  629400.493     13  48415.4225   Root MSE        =    176.86

------------------------------------------------------------------------------
gdp_percap~a | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
```

```
 cadre_ratio |   -347.3147     153.6286      -2.26     0.047      -689.6205      -5.00896
urban_employ |    1.227511     .6200836       1.98     0.076      -.1541212      2.609144
rural_employ |    .0931689     .1152433       0.81     0.438      -.1636091       .349947
       _cons |    145.9092     109.1341       1.34     0.211      -97.25676      389.0752
------------------------------------------------------------------------------
```

```
. outreg2 using regression_results.doc, append ctitle("M2: Control 1: Employment") dec(3)
regression_results.doc
dir : seeout
```

```
. reg gdp_percapita cadre_ratio industry_share

      Source |       SS       df       MS      Number of obs   =        27
-------------+------------------------------   F(2, 24)        =     35.47
       Model |  572790.163     2  286395.081   Prob > F        =    0.0000
    Residual |  193807.944    24  8075.33098   R-squared       =    0.7472
-------------+------------------------------   Adj R-squared   =    0.7261
       Total |  766598.106    26  29484.5425   Root MSE        =    89.863

------------------------------------------------------------------------------
 gdp_percapita | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
---------------+--------------------------------------------------------------
    cadre_ratio |  -148.1142   54.05246    -2.74   0.011    -259.673   -36.55544
industry_share |   969.7611   122.1487     7.94   0.000     717.6584    1221.864
         _cons |   -38.8702   52.05891    -0.75   0.463    -146.3145    68.5741
------------------------------------------------------------------------------
```

```
. outreg2 using regression_results.doc, append ctitle("M4: Control 2: Industry Share") dec(3)
regression_results.doc
dir : seeout
```

```
. reg l_gdp_percapita cadre_ratio industry_share

      Source |       SS       df       MS      Number of obs   =        27
-------------+------------------------------   F(2, 24)        =     32.24
       Model |  4.90959759     2   2.4547988   Prob > F        =    0.0000
    Residual |  1.82767305    24  .076153044   R-squared       =    0.7287
-------------+------------------------------   Adj R-squared   =    0.7061
       Total |  6.73727065    26  .259125794   Root MSE        =     .27596

------------------------------------------------------------------------------
l_gdp_percap~a | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
---------------+--------------------------------------------------------------
    cadre_ratio |  -.3510338   .1659888    -2.11   0.045    -.6936179   -.0084497
```

```
industry_share |    2.898342    .3751047       7.73    0.000       2.124164       3.67252
         _cons |    4.445404    .1598669      27.81    0.000       4.115455      4.775353
------------------------------------------------------------------------------

. outreg2 using regression_results.doc, append ctitle("M5: Log GDP + Industry Share") dec(3)
regression_results.doc
dir : seeout

. reg l_gdp_percapita cadre_ratio

      Source |       SS       df       MS      Number of obs   =        27
-------------+------------------------------   F(1, 25)        =      1.42
       Model |  .363052784     1  .363052784   Prob > F        =    0.2440
    Residual |  6.37421786    25  .254968715   R-squared       =    0.0539
-------------+------------------------------   Adj R-squared   =    0.0160
       Total |  6.73727065    26  .259125794   Root MSE        =     .50494

------------------------------------------------------------------------------
l_gdp_perc~a | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
 cadre_ratio |  -.3624124   .3037118    -1.19   0.244    -.9879185    .2630937
       _cons |   5.462615   .1659591    32.92   0.000     5.120816    5.804414
------------------------------------------------------------------------------

. outreg2 using regression_results.doc, append ctitle("M6: Log Baseline") dec(3)
regression_results.doc
dir : seeout

. reg l_gdp_percapita cadre_ratio urban_employ rural_employ

      Source |       SS       df       MS      Number of obs   =        14
-------------+------------------------------   F(3, 10)        =      3.05
       Model |  2.45278052     3  .817593506   Prob > F        =    0.0788
    Residual |  2.68069068    10  .268069068   R-squared       =    0.4778
-------------+------------------------------   Adj R-squared   =    0.3211
       Total |   5.1334712    13   .3948824   Root MSE        =     .51775

------------------------------------------------------------------------------
l_gdp_perc~a | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
 cadre_ratio |  -1.000246   .4497422    -2.22   0.050    -2.002334     .001842
urban_employ |    .004592   .0018153     2.53   0.030     .0005473    .0086367
rural_employ |  -.0001441   .0003374    -0.43   0.678    -.0008958    .0006076
       _cons |   5.067582   .3194864    15.86   0.000     4.355722    5.779442
```

```
-------------------------------------------------------------------------------

. outreg2 using regression_results.doc, append ctitle("M7: Log GDP + Employment") dec(3)
regression_results.doc
dir : seeout

. twoway (scatter gdp_percapita cadre_ratio) ///
/ / / is not a twoway plot type
r(198);

.           (lfit gdp_percapita cadre_ratio), ///
( is not a valid command name
r(199);

.           title("Cadre Ratio vs GDP per Capita") ///
command title is unrecognized
r(199);

.           xtitle("Cadre Ratio") ///
command xtitle is unrecognized
r(199);

.           ytitle("GDP per Capita")
command ytitle is unrecognized
r(199);

. twoway (scatter gdp_percapita cadre_ratio) (lfit gdp_percapita cadre_ratio),title("Cadre Ratio
vs GDP per
>   Capita") xtitle("Cadre Ratio") ytitle("GDP per Capita")

. graph save "Graph" "/Users/liujiatong/Desktop/ESPS0015 final/M1 scatter fi.gph"
file /Users/liujiatong/Desktop/ESPS0015 final/M1 scatter fi.gph saved

. graph save "Graph" "/Users/liujiatong/Desktop/ESPS0015 final/M1 scatter fi.gph", replace
file /Users/liujiatong/Desktop/ESPS0015 final/M1 scatter fi.gph saved

. graph save "Graph" "/Users/liujiatong/Desktop/ESPS0015 final/M1 scatter fi.gph", replace
file /Users/liujiatong/Desktop/ESPS0015 final/M1 scatter fi.gph saved

. graph use "/Users/liujiatong/Desktop/ESPS0015 final/M1 scatter fi.gph"

. graph export "/Users/liujiatong/Desktop/ESPS0015 final/M1 scatter fit.png", as(png)
name("Graph")
file /Users/liujiatong/Desktop/ESPS0015 final/M1 scatter fit.png saved as PNG format
```

```
. reg l_gdp_percapita cadre_ratio industry_share

      Source |       SS       df       MS      Number of obs   =        27
-------------+------------------------------   F(2, 24)        =     32.24
       Model |  4.90959759     2  2.4547988   Prob > F        =    0.0000
    Residual |  1.82767305    24  .076153044   R-squared       =    0.7287
-------------+------------------------------   Adj R-squared   =    0.7061
       Total |  6.73727065    26  .259125794   Root MSE        =    .27596

------------------------------------------------------------------------------
l_gdp_percap~a | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
---------------+--------------------------------------------------------------
   cadre_ratio |  -.3510338   .1659888    -2.11   0.045    -.6936179   -.0084497
industry_share |   2.898342   .3751047     7.73   0.000     2.124164    3.67252
         _cons |   4.445404   .1598669    27.81   0.000     4.115455    4.775353
------------------------------------------------------------------------------

. margins, at(cadre_ratio=(0(.1)1))

Predictive margins                                      Number of obs = 27
Model VCE: OLS

Expression: Linear prediction, predict()
1._at:  cadre_ratio =  0
2._at:  cadre_ratio = .1
3._at:  cadre_ratio = .2
4._at:  cadre_ratio = .3
5._at:  cadre_ratio = .4
6._at:  cadre_ratio = .5
7._at:  cadre_ratio = .6
8._at:  cadre_ratio = .7
9._at:  cadre_ratio = .8
10._at: cadre_ratio = .9
11._at: cadre_ratio =  1

------------------------------------------------------------------------------
             |            Delta-method
             |     Margin   std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         _at |
          1  |   5.457574   .0907011    60.17   0.000     5.270377    5.644772
          2  |   5.422471   .0778542    69.65   0.000     5.261788    5.583154
          3  |   5.387368   .0666853    80.79   0.000     5.249736    5.524999
```

```
           4  |   5.352264   .0581688    92.01   0.000      5.23221     5.472319
           5  |   5.317161   .0535849    99.23   0.000     5.206567     5.427755
           6  |   5.282058   .0539455    97.91   0.000      5.17072     5.393396
           7  |   5.246954   .0591603    88.69   0.000     5.124853     5.369055
           8  |   5.211851   .0681235    76.51   0.000     5.071251     5.352451
           9  |   5.176747   .0795784    65.05   0.000     5.012506     5.340989
          10  |   5.141644    .092605    55.52   0.000     4.950517     5.332771
          11  |   5.106541   .1066288    47.89   0.000      4.88647     5.326612
------------------------------------------------------------------------------
```

. marginsplot, title("Marginal Effect of Cadre Ratio on Log GDP per Capita")

Variables that uniquely identify margins: cadre_ratio

. graph export "/Users/liujiatong/Desktop/ESPS0015 final/M5 Margin.png", as(png) name("Graph")
file /Users/liujiatong/Desktop/ESPS0015 final/M5 Margin.png saved as PNG format

. twoway (scatter l_gdp_percapita cadre_ratio) (lfit l_gdp_percapita cadre_ratio),title("Cadre Ratio vs Log
> GDP per Capita (Baseline)") xtitle("Cadre Ratio") ytitle("Log GDP per Capita")

. graph export "/Users/liujiatong/Desktop/ESPS0015 final/M6 log base scatter fit.png", as(png) name("Graph"
>)
file /Users/liujiatong/Desktop/ESPS0015 final/M6 log base scatter fit.png saved as PNG format

. eststo M5: reg l_gdp_percapita cadre_ratio industry_share

```
      Source |       SS           df       MS      Number of obs   =        27
-------------+----------------------------------   F(2, 24)        =     32.24
       Model |  4.90959759         2  2.4547988   Prob > F        =    0.0000
    Residual |  1.82767305        24  .076153044   R-squared       =    0.7287
-------------+----------------------------------   Adj R-squared   =    0.7061
       Total |  6.73727065        26  .259125794   Root MSE        =    .27596

------------------------------------------------------------------------------
l_gdp_percap~a | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
---------------+--------------------------------------------------------------
   cadre_ratio |  -.3510338   .1659888    -2.11   0.045    -.6936179   -.0084497
industry_share |   2.898342   .3751047     7.73   0.000     2.124164    3.67252
         _cons |   4.445404   .1598669    27.81   0.000     4.115455    4.775353
------------------------------------------------------------------------------
```

```
. eststo M6: reg l_gdp_percapita cadre_ratio

      Source |       SS       df       MS      Number of obs   =        27
-------------+------------------------------   F(1, 25)        =      1.42
       Model |  .363052784     1  .363052784   Prob > F        =    0.2440
    Residual |  6.37421786    25  .254968715   R-squared       =    0.0539
-------------+------------------------------   Adj R-squared   =    0.0160
       Total |  6.73727065    26  .259125794   Root MSE        =    .50494

------------------------------------------------------------------------------
l_gdp_perc~a | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
 cadre_ratio |  -.3624124   .3037118    -1.19   0.244    -.9879185    .2630937
       _cons |   5.462615   .1659591    32.92   0.000     5.120816    5.804414
------------------------------------------------------------------------------

. eststo M7: reg l_gdp_percapita cadre_ratio urban_employ rural_employ

      Source |       SS       df       MS      Number of obs   =        14
-------------+------------------------------   F(3, 10)        =      3.05
       Model |  2.45278052     3  .817593506   Prob > F        =    0.0788
    Residual |  2.68069068    10  .268069068   R-squared       =    0.4778
-------------+------------------------------   Adj R-squared   =    0.3211
       Total |   5.1334712    13   .3948824   Root MSE        =    .51775

------------------------------------------------------------------------------
l_gdp_perc~a | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
 cadre_ratio |  -1.000246   .4497422    -2.22   0.050    -2.002334    .001842
urban_employ |    .004592   .0018153     2.53   0.030     .0005473    .0086367
rural_employ |  -.0001441   .0003374    -0.43   0.678    -.0008958    .0006076
       _cons |   5.067582   .3194864    15.86   0.000     4.355722    5.779442
------------------------------------------------------------------------------

. ssc install coefplot
checking coefplot consistency and verifying not already installed...
installing into /Users/liujiatong/Library/Application Support/Stata/ado/plus/...
installation complete.

. coefplot M5 M6 M7, keep(cadre_ratio) xline(0) title("Effect of Cadre Ratio on Log GDP per
Capita") legend
> (label(1 "M5") label(2 "M6") label(3 "M7"))

.
```

. graph export "/Users/liujiatong/Desktop/ESPS0015 final/M7 coefplot.png", as(png) name("Graph")
file /Users/liujiatong/Desktop/ESPS0015 final/M7 coefplot.png saved as PNG format